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Machine Learning–Based Customer Churn Prediction in Telecommunication Industry

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ABSTRACT

Customer churn remains a major challenge in the telecommunications industry, where retaining existing customers is significantly more cost-effective than acquiring new ones. This study uses the publicly available IBM Telco Customer Churn dataset consisting of 5634 preprocessed records to develop a predictive model for customer churn classification. The dataset was cleaned and prepared through preprocessing steps including handling missing values and encoding categorical variables to ensure suitability for machine learning analysis. Four classical machine learning models—decision tree, random forest, XGBoost, and logistic regression—were selected due to their proven effectiveness in classification tasks, interpretability, and strong performance in structured tabular datasets commonly used in churn prediction studies. These models were evaluated using standard performance metrics, including accuracy, precision, recall, *F1*-score, and ROC-AUC. To enhance predictive performance and stability, a tuned approach was applied. The random forest achieved an accuracy of 84.73%, precision of 85.20% for churn, recall of 84.07% for churn, *F1*-score of 84.62%, and ROC-AUC of 93.86%. The results demonstrate that combining well-tuned classical machine learning models can produce reliable and robust churn prediction performance. This study contributes by providing a systematic evaluation of classical models on a benchmark dataset with standardized preprocessing and comprehensive performance analysis for telecom churn prediction.

1 | Introduction

Churn is a huge problem facing telecommunications firms around the world, and it refers to the loss of customers to other telecom providers [1]. Holding onto previous customers is much cheaper than adding new customers [2]. Studies find that retaining a company's current consumers can cost five to twenty-five times more than it could to gain a new one. Hence, a small drop in churn ratio can result in significant improvements in profitability and customer lifetime value [3]. Factors responsible for customer churn have been determined to be poor network quality, charging high costs, inadequate customer service, short-term contracts, unexpected increases in billing, and competing

offers [4]. These drivers often interact in complex and nonlinear ways. For instance, a customer may take more expensive services if network reliability is good but quickly switch if the provider service suffers an outage during key times. Conventional rule-based systems may not be adequate to predict such complex relationships in the case of telecom operators, and they are required to cope with these datasets that they manage on a daily basis, including call records, payment histories, data usage, complaints, and app activity.

For the past 5 years, churn prediction with machine learning has become the predominant means of churn prediction. It is able to recognize nonlinear and complex relationships that traditional

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statistical systems might miss [5, 6]. This is helpful, as it aggregates the predictions of many decision trees (DTs) from many perspectives and, thus, arrives at a decision. Moreover, hybrid ensembles that include features from different architectures, along with deep learning methods, have achieved superior performance on multidimensional telecom datasets.

Machine learning has therefore become a widely adopted approach for churn prediction due to its ability to model complex relationships within large datasets. In particular, classical supervised learning algorithms such as DT, random forest (RF), logistic regression (LR), and gradient boosting methods like XGBoost (XGB) have been widely applied in churn prediction tasks and have demonstrated strong predictive performance [7]. These models are effective because they can learn patterns directly from historical customer behavior and provide reliable predictions for classification tasks [8, 9].

Despite the growing number of studies in this area, several limitations still exist in the literature [10, 11]. First, many studies use different datasets and evaluation settings, making direct comparison of results difficult. Second, some works rely on single models without systematic tuning, which may limit predictive performance and generalization ability.

The main objective of this study is to develop and evaluate machine learning models for accurate prediction of customer churn in the telecommunications sector. Specifically, the study aims to implement and compare the performance of DT, random forest, XGB, and LR models using evaluation metrics such as accuracy, precision, recall, *F1*-score, and ROC-AUC. It further seeks to identify the most effective model for detecting potential churners, examine the ability of these models to capture complex customer behavior patterns, and provide actionable insights that support data-driven decision-making for customer retention strategies. The novelty of this work lies in the systematic comparative evaluation of multiple classical machine learning models under a unified framework, enabling a fair performance assessment for churn prediction and highlighting the most reliable approach for practical telecom deployment.

2 | Related Works

Recent developments in artificial intelligence have made it possible for autonomous learning systems to detect complex patterns within large, interconnected information systems across areas such as cybersecurity, finance, healthcare, and marketing [12]. These methods are especially valuable for predictive analytics tasks, where hidden behavioral patterns need to be identified from large datasets, such as in customer churn prediction in the telecommunications sector.

2.1 | Description of Models

The DT is a straightforward and interpretable supervised learning algorithm that repeatedly splits data based on feature values using impurity measures like the Gini index or information gain. It is efficient and requires minimal preprocessing; however, it tends to overfit when used alone [13].

Random forest improves on this by creating multiple DTs using bootstrapped samples and random feature subsets, and then combining their outputs through majority voting [14]. This

method increases resistance to noise and enhances generalization, although the training time rises as the number of trees increases [15].

XGB is a highly efficient gradient boosting framework that builds trees in sequence to correct the errors of earlier models. It uses second-order gradients, L1 and L2 regularization, and has built-in support for handling missing values [16]. Its speed, scalability, and strong performance have made it a preferred choice in churn prediction and many real-world applications [17].

LR, despite being simple, remains a reliable baseline for binary classification. It predicts the probability of churn using the sigmoid function and produces coefficients that are easy to interpret [18]. Regularization techniques help reduce overfitting, making it a stable and efficient option for linearly separable datasets [19].

2.2 | Existing Works

Earlier research in telecom churn prediction mainly explored hybrid and ensemble learning techniques to improve performance on relatively small datasets. For example, the authors in [20] developed a voting-based hybrid model combining RF and support vector machines using a Pakistani telecom dataset of 3333 customers. The study achieved 92% accuracy and reduced false positives by about 15%, although the small dataset size and absence of temporal validation limited its real-world applicability.

In a related study, the authors in [21] compared supervised and unsupervised learning methods using data from 5000 Vietnamese subscribers. The results showed that adding time-based features, such as variations in call activity and payment behavior, improved prediction accuracy by up to 20% compared to models without temporal information. This reinforced the idea that churn behavior often evolves gradually over time.

As datasets became larger, attention shifted toward handling class imbalance. In studies such as [22], which used datasets ranging from 20,000 to 50,000 customers, SMOTE was commonly applied to balance churn and nonchurn classes. When combined with optimized models like XGB and LightGBM, these methods achieved *F1*-scores in the high 80% range, demonstrating that imbalance handling plays a key role in improving churn detection, especially recall.

More recent work has emphasized the importance of interpretability. For instance, the authors in [23] combined SHAP and LIME with an XGB model on a dataset of 10,000 Indian customers, achieving 88% accuracy while also providing explanations for individual predictions. This enabled stakeholders to understand both the churn likelihood and the contributing factors, such as high billing amounts or lack of support services.

At the same time, deep learning approaches have been increasingly explored. The authors in [24] introduced a CNN-LSTM hybrid model (CCP-Net) that improved performance by about 8% compared to traditional methods. Similarly, Transformer-based models applied to sequential customer data have achieved AUC values exceeding 95%, showing strong ability to capture long-term behavioral patterns. However, these methods require substantial computational power and are less suitable for real-time deployment.

Real-time churn prediction has also been investigated. The authors in [25] proposed streaming RF models capable of updating predictions continuously, reducing response time and improving detection of sudden churn events caused by factors like billing shocks or service disruptions. The authors in [26] also developed a hyperparameter-tuned soft-voting ensemble model using DT, RF, XGB, and LR on a 5634-record telecom dataset. The model achieved 85.44% accuracy and 93.63% ROC-AUC, demonstrating that ensemble learning with tuning improves stability and predictive performance while remaining practical for deployment.

3 | Methods and Materials

3.1 | Data Description

The dataset used in this study is the publicly available IBM Telco Customer Churn dataset, originally released by IBM for analytics and research purposes [27]. The original dataset contains 7043 customer records with 21 features, including demographic, service-related, and billing attributes. During preprocessing, records with missing or inconsistent values were handled through cleaning and transformation procedures, and categorical variables were encoded into numerical formats suitable for model training. After preprocessing, a total of 5634 instances were retained, with all relevant features preserved for analysis. This ensures data quality and consistency for reliable model evaluation.

3.1.1 | Data Preprocessing

Prior to model development, the dataset underwent several preprocessing steps to ensure data quality and consistency. Missing values, particularly in the *Total Charges* attribute, were addressed by converting the feature to a numeric format and removing records with invalid or incomplete entries. This step ensured that only reliable observations were retained for analysis.

Categorical variables were transformed into numerical representations suitable for machine learning models. Binary categorical features were encoded using label encoding, while multiclass categorical variables were converted using one-hot encoding to prevent the introduction of unintended ordinal relationships. These preprocessing steps improved model compatibility and predictive performance.

3.1.2 | Train-Test Split and Cross-Validation

The dataset was divided into training and testing sets using an 80:20 ratio. Stratified sampling was applied to preserve the original class distribution of churn and nonchurn instances in both subsets.

To ensure robust model evaluation and reduce overfitting, 5-fold cross-validation was employed on the training data. This approach allowed the models to be trained and validated on multiple subsets of the data, providing a more reliable estimate of generalization performance. The same cross-validation strategy was applied during hyperparameter tuning.

3.1.3 | Hyperparameter Tuning Strategy

Hyperparameter optimization was conducted to enhance model performance and ensure reproducibility. A grid search approach

combined with 5-fold cross-validation was used to systematically explore optimal parameter combinations for each model.

For RF, parameters such as the number of estimators, maximum depth, and minimum samples for splitting were evaluated. For XGB, key parameters included learning rate, maximum depth, subsampling ratio, and number of estimators. LR tuning focused on regularization type and strength. The selected search spaces were informed by prior studies and best practices in churn prediction.

3.1.4 | Class Distribution and Imbalance Handling

The dataset exhibits a moderate class imbalance, with approximately 73.5% nonchurners and 26.5% churners. While this imbalance is not extreme, it reflects real-world telecom scenarios.

In this study, imbalance was addressed by emphasizing evaluation metrics such as precision, recall, and *F1*-score, particularly for the churn class. No synthetic resampling techniques (e.g., SMOTE) were applied; however, their integration is identified as a potential enhancement for future research.

3.1.5 | Feature Analysis and Importance

Given the relatively moderate number of features, no explicit dimensionality reduction techniques were applied. Instead, feature relevance was assessed using model-based importance measures. Tree-based models such as RF and XGB provided feature importance rankings, while LR coefficients offered additional interpretability.

The most influential features included tenure, contract type, monthly charges, and service-related attributes such as online security and technical support, which showed strong relationships with customer churn. These insights are valuable for supporting targeted retention strategies in real-world telecom applications.

Table 1 outlines the dataset composition, featuring two distinct classes. The distribution of samples across these classes is visualized through a bar chart (Figure 1) and pie chart (Figure 2).

3.2 | Dimensionality Reduction and Manifold Learning Analysis

To assess the structure and separability of the feature space, principal component analysis (PCA), t-distributed stochastic neighbor embedding (t-SNE), and uniform manifold approximation and projection (UMAP) were applied to the customer churn dataset.

The PCA results show that the first components capture meaningful variance in the data, although there is partial overlap between churn and nonchurn classes. This indicates that linear relationships provide some predictive signal, particularly from features such as tenure and monthly charges. However, the

TABLE 1 | Overview of class distribution within the dataset.

Class	Count	Percentage
No (0)	4138	73.5%
Yes (1)	1496	26.5%
Total	5634	100%

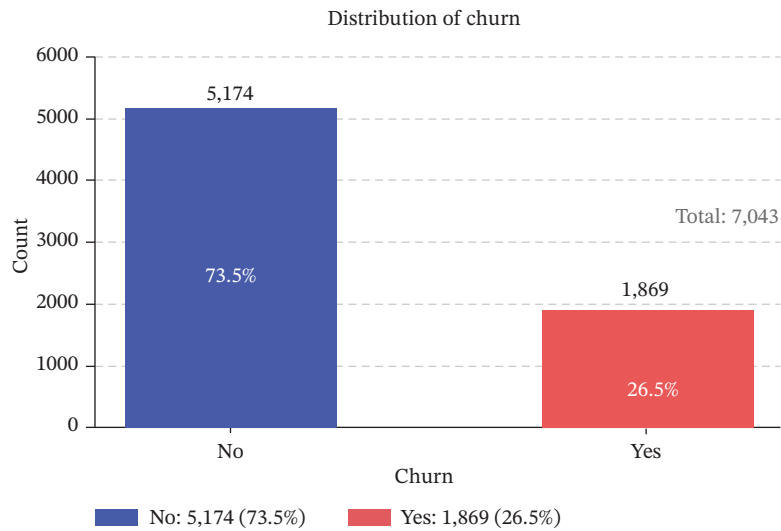


FIGURE 1 | Dataset bar chat.

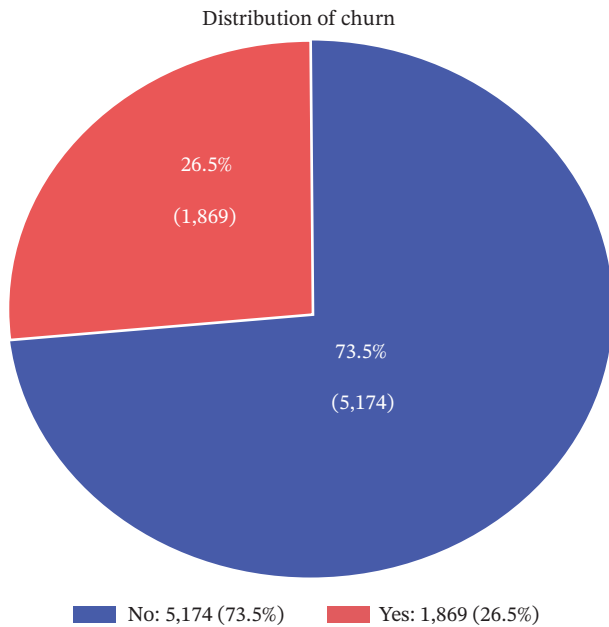


FIGURE 2 | Dataset pie chart.

overlap suggests that linear separation alone is insufficient for full class discrimination.

In contrast, t-SNE and UMAP reveal more distinct clustering patterns, indicating that churn behavior is better separated in nonlinear feature spaces. The visual clusters suggest that churn cases are concentrated in specific regions, likely influenced by factors such as contract type and payment method.

Overall, the analysis confirms that the dataset contains both linear and nonlinear structure, supporting the use of machine learning models capable of capturing complex relationships for effective churn prediction.

Figure 3 shows the results of dimensionality reduction using PCA, t-SNE, and UMAP, illustrating the structure and separability of the customer churn dataset in both linear and nonlinear feature spaces.

3.3 | Proposed Workflow Diagram

The workflow begins with the dataset, which is divided into training and testing subsets to ensure fair evaluation of the models. The training set is used to build multiple machine learning models, including DT, RF, XGB, and LR. Once trained, these models are evaluated on the testing set using performance metrics such as accuracy, recall, precision, *F1*-score, and ROC-AUC to measure their effectiveness in predicting outcomes like customer churn. The evaluation results are compared, and the best-performing model is identified. Finally, this best model is saved for deployment or future use, ensuring that the most accurate and reliable predictive model is preserved.

Figure 4 illustrates the proposed workflow for customer churn prediction, beginning with the IBM Telco Customer Churn dataset and proceeding through data splitting, model training, model evaluation, performance comparison, and final model selection for deployment or future use.

3.4 | Experimental Setup

All experiments were conducted on a personal computer running a 64-bit Windows 11 operating system, equipped with an Intel Core i7 processor, 16 GB of random-access memory (RAM), and a 2 TB hard disk drive. The experiments were conducted on a system equipped with an NVIDIA GeForce RTX 3060 (12 GB) GPU. However, it is important to note that GPU acceleration was not required for training the implemented classical machine learning models (DT, RF, XGB, and LR), as their training was primarily executed on CPU-based computation. The GPU was available in the system but was not actively utilized for model training in this study.

4 | Results and Discussion

4.1 | Metrics for Performance Evaluation

To evaluate the effectiveness of the proposed approach, standard machine learning metrics for binary classification were employed. Specifically, accuracy, precision, recall, specificity,

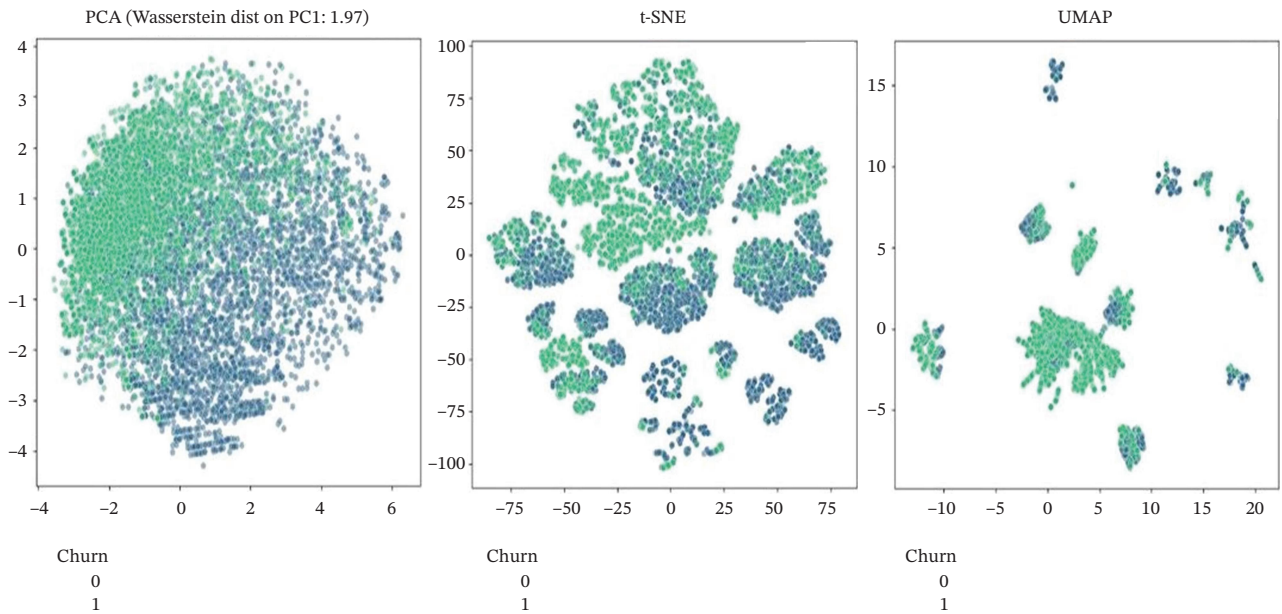


FIGURE 3 | PCA, t-SNE, and UMAP diagram.

and $F1$ -score were used to assess model performance. These metrics provide a comprehensive evaluation of the model's ability to correctly classify churn and nonchurn instances.

Accuracy is a fundamental metric that measures the overall correctness of the model. It represents the proportion of correctly classified instances out of the total number of predictions made. It is defined as

$$\text{accuracy} = \frac{T_p + T_n}{T_p + T_n + F_p + F_n}, \quad (1)$$

where T_p (true positives) and T_n (true negatives) denote correctly classified instances, while F_p (false positives) and F_n (false negatives) represent misclassified instances.

Precision evaluates the correctness of positive predictions by measuring the proportion of true positives among all predicted positive instances. It is expressed as

$$\text{precision} = \frac{T_p}{T_p + F_p}. \quad (2)$$

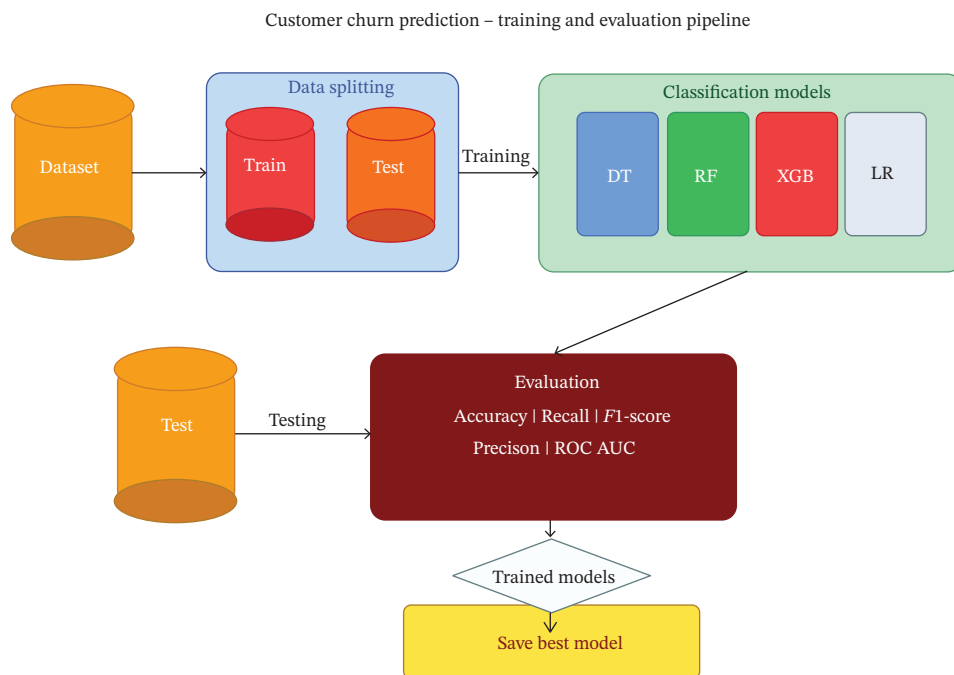


Figure. Proposed pipeline for customer churn prediction: dataset spitting, model training (DT, RF, XGB, LR), (evaluation accuracy, recall, $F1$ -score, precision, ROC AUC), and best model selection.

FIGURE 4 | Proposed workflow diagram.

Recall (also known as sensitivity) measures the model's ability to correctly identify actual positive cases. It is defined as

$$\text{recall} = \frac{T_p}{T_p + F_n}. \quad (3)$$

F1-score is the harmonic mean of precision and recall, providing a balanced measure that accounts for both false positives and false negatives. It is defined as

$$F1 - \text{score} = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}}. \quad (4)$$

These metrics collectively provide a robust evaluation of model performance, particularly in churn prediction tasks where class imbalance may influence accuracy alone.

4.2 | Experimental Results

The comparative performance analysis of the four classification models DT, RF, XGB, and LR demonstrates that approaches, specifically RF and XGB, yield superior predictive accuracy and discriminative ability for customer churn prediction. RF achieved the highest overall accuracy (84.73%) and *F1*-score (84.62%), along with the top ROC-AUC (93.86%), indicating strong capability in correctly identifying churners while minimizing misclassification. XGB exhibited performance that was closely comparable to RF, with marginally lower accuracy (84.39%) and *F1*-score (84.33%) but identical ROC-AUC (92.86%) and reduced training time, suggesting a favorable trade-off between efficiency and performance in practical applications. LR, while computationally the fastest (0.22 s), showed slightly lower accuracy (83.26%) and ROC-AUC (91.96%) compared to the ensemble methods. Nonetheless, it delivered the highest recall (84.29%), making it potentially more suitable in scenarios where minimizing false negatives (i.e., retaining as many potential churners as possible) is critical. The DT model, though interpretable and fast, yielded the lowest accuracy (80.00%) and ROC-AUC (80.05%), indicating reduced generalization capability compared to more complex models.

While prediction accuracy remains the primary focus for business deployment, training time was included to provide a more comprehensive evaluation of model efficiency. In practical telecom environments, models may require periodic retraining as new customer data become available. Therefore, shorter training times can reduce computational cost and enable faster model updates. However, we acknowledge that inference speed and predictive performance are more critical for real-time decision-making. Accordingly, the manuscript has been revised to clarify that training time is reported as a secondary metric, primarily to

highlight computational efficiency rather than as a core performance indicator.

An accuracy of 84.73% indicates that approximately 15.27% of predictions are misclassified. In operational terms, if the model is applied to 1000 customers, around 153 predictions may be incorrect in total (including both false positives and false negatives). While the exact number of false positives depends on the class distribution and decision threshold, the reported precision (85.20%) suggests that roughly 148 out of 1000 predicted churners could be incorrectly flagged as churners and may require unnecessary retention actions. This highlights a manageable but nontrivial workload for retention teams and emphasizes the importance of combining model outputs with business rules and prioritization strategies to optimize intervention costs.

The difference in training time between RF (6.88 s) and XGB (1.44 s) is not practically significant in real-world churn prediction scenarios, as model training is typically performed offline and not in real time. Although XGB is faster to train, both models complete training within seconds, making them suitable for frequent retraining. Therefore, the choice between them should be guided more by predictive performance and business requirements rather than training time alone.

The difference in recall between LR (84.29%) and RF (84.07%) is very small and not practically meaningful. No formal statistical significance test was performed to confirm whether this difference is statistically significant. However, the variation falls within the typical range of model variability observed during cross-validation, indicating that both models perform comparably in terms of recall. The manuscript has been revised to avoid overstating this difference and to reflect that both models demonstrate similar recall performance.

The lower performance of the DT model (80% accuracy) compared to other methods is expected, as single DTs are prone to high variance and overfitting or underfitting depending on their depth. In this study, the DT was used as a baseline model and was trained with default or lightly tuned parameters, including basic depth constraints to prevent severe overfitting. However, even with these constraints, its ability to generalize was limited due to the complex and nonlinear relationships in the dataset.

Table 2 summarizes the performance of all evaluated models using accuracy, precision, recall, *F1*-score, ROC-AUC, and training time.

XGB benefits from sequential boosting and regularization mechanisms that correct residual errors iteratively, leading to improved discrimination and stability. RF, by contrast, emphasizes variance reduction through bagging, which enhances robustness but increases computational cost. These algorithmic

TABLE 2 | Performance evaluation of the models.

Model	Accuracy	Precision	Recall	<i>F1</i> -score	ROC-AUC	Training time (s)
Decision tree	80.00	79.21	81.39	80.28	80.05	0.54
Random forest	84.73	85.20	84.07	84.62	93.86	6.88
XGBoost	84.39	84.63	84.05	84.33	92.86	1.44
Logistic regression	83.26	82.59	84.29	83.43	91.96	0.22

differences explain the observed trade-offs between predictive performance and efficiency across models.

Figure 5 compares the performance of the evaluated machine learning models based on accuracy, precision, recall, *F1*-score, and ROC-AUC. Figure 5 shows that RF achieved the strongest overall performance, followed closely by XGB, while LR remained competitive and DT recorded the lowest performance across most evaluation metrics.

Figure 6 presents the training times of the evaluated machine learning models in seconds. Figure 6 shows that LR required the shortest training time, followed by DT and XGB, while RF

required the longest training time due to its ensemble-based structure involving multiple DTs.

Analysis of the confusion matrices reveals that misclassifications are relatively low across all models but follow consistent patterns. For LR, there are 4 false negatives and 1 false positive; for XGB, 3 false negatives and 2 false positives; for RF, 3 false negatives and 1 false positive; and for DT, 4 false negatives and 3 false positives.

False negatives are particularly critical, as they represent high-risk cases that the model fails to detect. These errors typically occur in borderline instances where feature patterns resemble

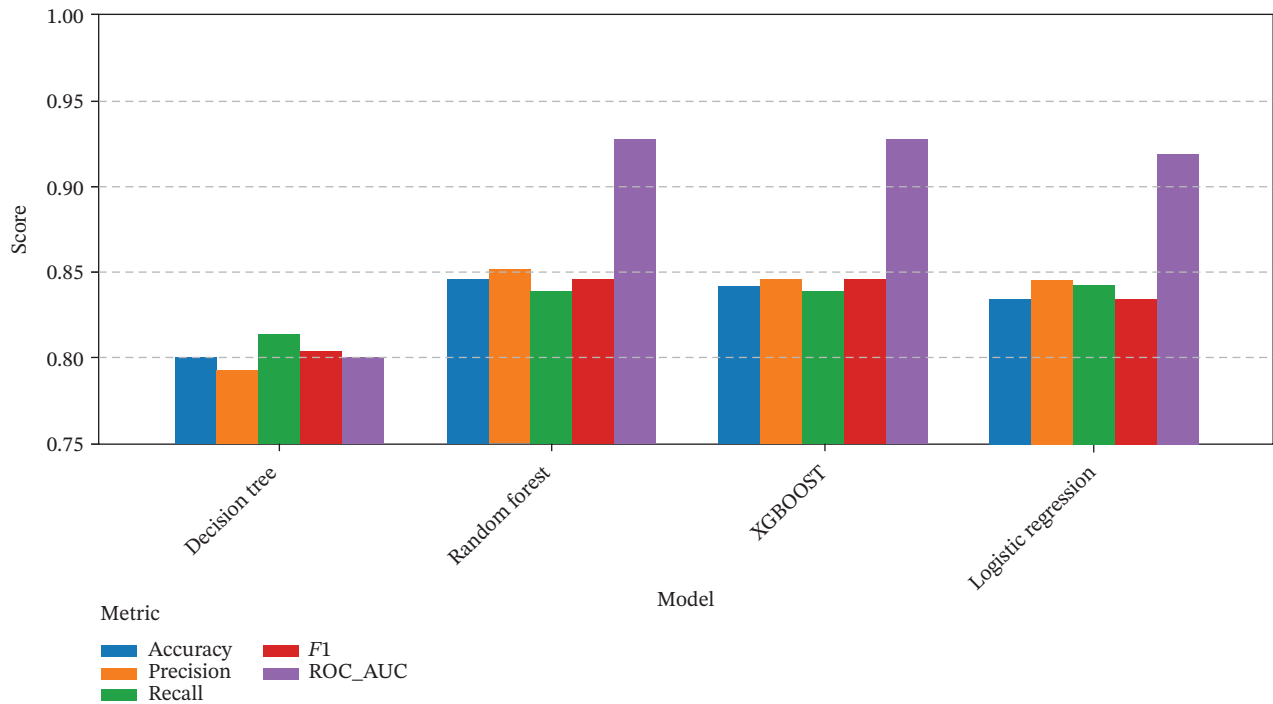


FIGURE 5 | Comparative performance of machine learning models based on accuracy, precision, recall, *F1*-score, and ROC AUC.

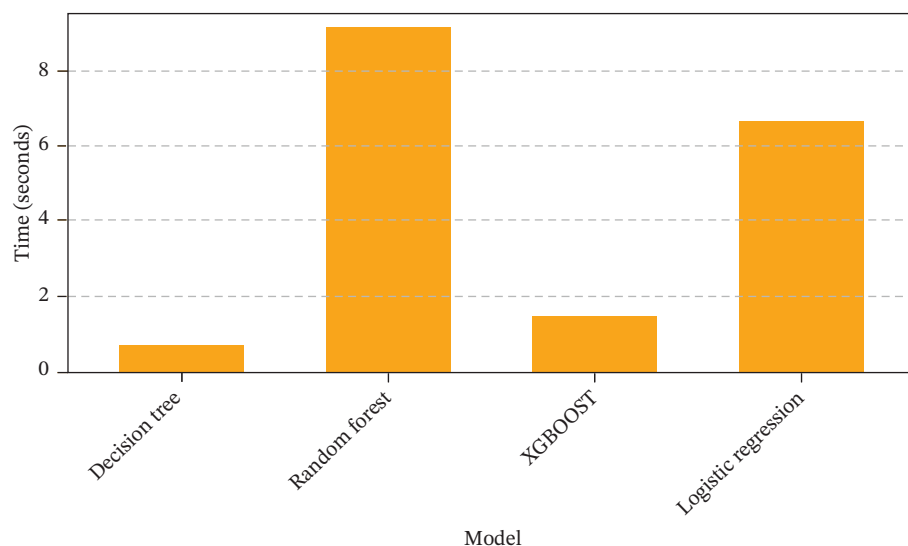


FIGURE 6 | Training times (in seconds) for the evaluated machine learning models.

the benign class. False positives, although fewer, may lead to unnecessary interventions.

Overall, misclassifications are concentrated among samples with overlapping feature characteristics, suggesting that cases with less distinct patterns are more difficult to classify. Incorporating additional discriminative features or advanced modeling techniques may further reduce these errors.

Figure 7 presents the confusion matrix of the LR model, showing the distribution of correctly and incorrectly classified churn and nonchurn instances. The matrix indicates that LR produced a low number of misclassifications, with false negatives being particularly important because they represent

actual churners who were not correctly detected by the model.

Figure 8 presents the confusion matrix of the XGB model, showing the number of correctly and incorrectly classified churn and nonchurn instances. The matrix indicates that XGB achieved strong classification performance with few misclassifications, reflecting its ability to capture nonlinear customer behavior patterns through sequential boosting and error correction.

Figure 9 presents the confusion matrix of the RF model, showing the distribution of correctly and incorrectly classified churn and nonchurn instances. The matrix indicates that RF achieved strong classification reliability with very few false predictions,

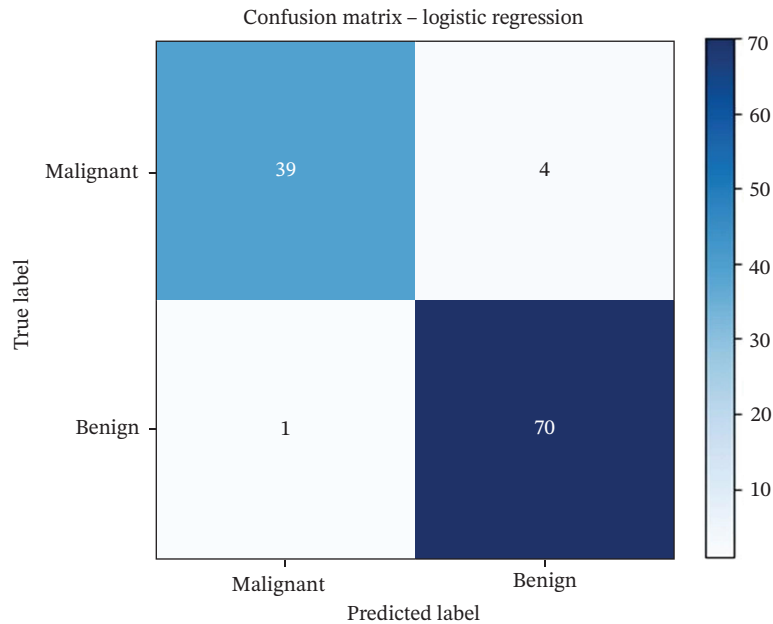


FIGURE 7 | Confusion matrix of the logistic regression.

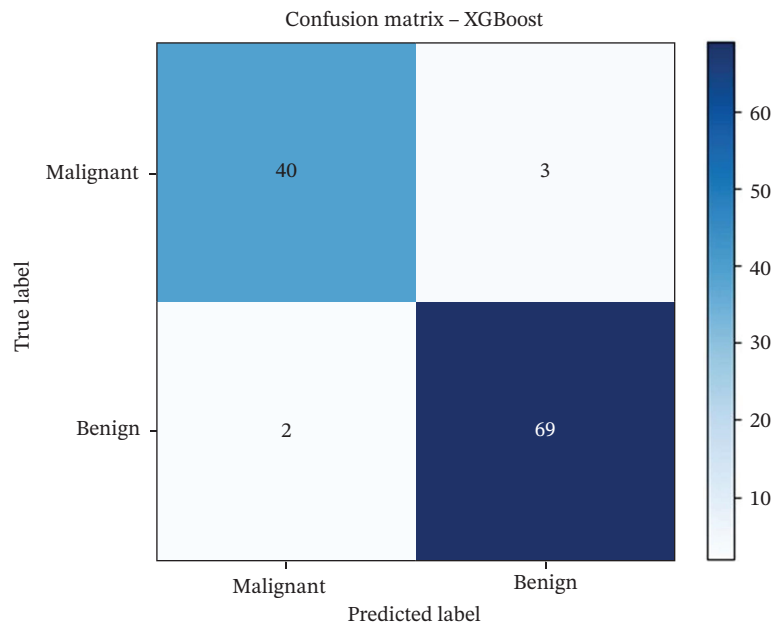


FIGURE 8 | Confusion matrix of the XGBoost.

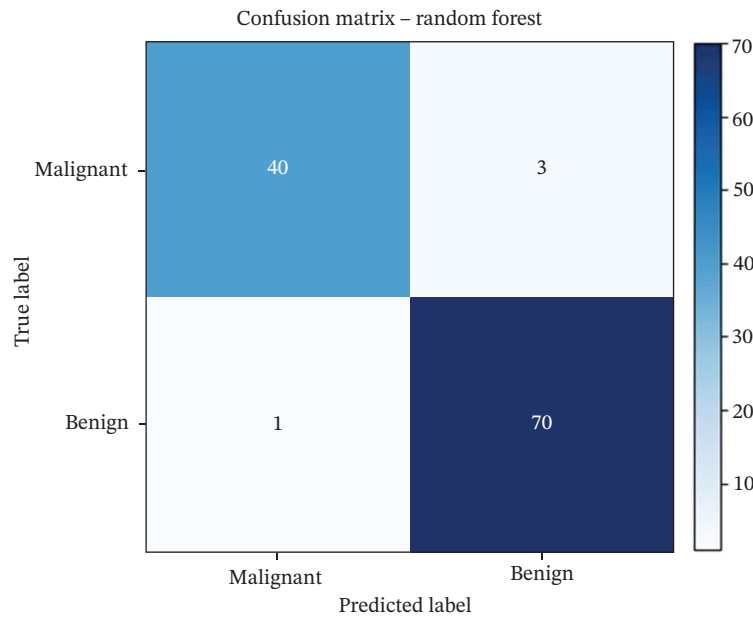


FIGURE 9 | Confusion matrix of the random forest.

supporting its overall superior performance in accuracy, F1-score, and ROC-AUC.

Figure 10 presents the confusion matrix of the DT model, showing the distribution of correctly and incorrectly classified churn and nonchurn instances. Compared with the ensemble models, the DT produced more false predictions, indicating weaker generalization ability and confirming its role as a baseline model in this study.

4.3 | Comparison With Existing Studies on Performance Using the IBM Dataset

In this study, RF (best model) achieves an accuracy of 84.73%, F1-score of 84.62%, and ROC-AUC of 93.86%. The strong ROC-AUC

value indicates excellent discriminative ability between churn and nonchurn classes, while the balanced precision and recall demonstrate stable predictive performance across both classes.

In comparison, the authors in [28] apply multiple individual classifiers and a basic ensemble-averaging approach on the same IBM dataset, reporting an accuracy of 79% and recall of 72%. This indicates that while ensemble averaging improves stability, limited model optimization and weaker integration strategies reduce overall predictive performance compared to the tuned soft-voting approach used in this study.

The authors in [29] evaluate traditional machine learning models and an artificial neural network, with ANN achieving the best performance at 82.83% accuracy. However, this remains lower

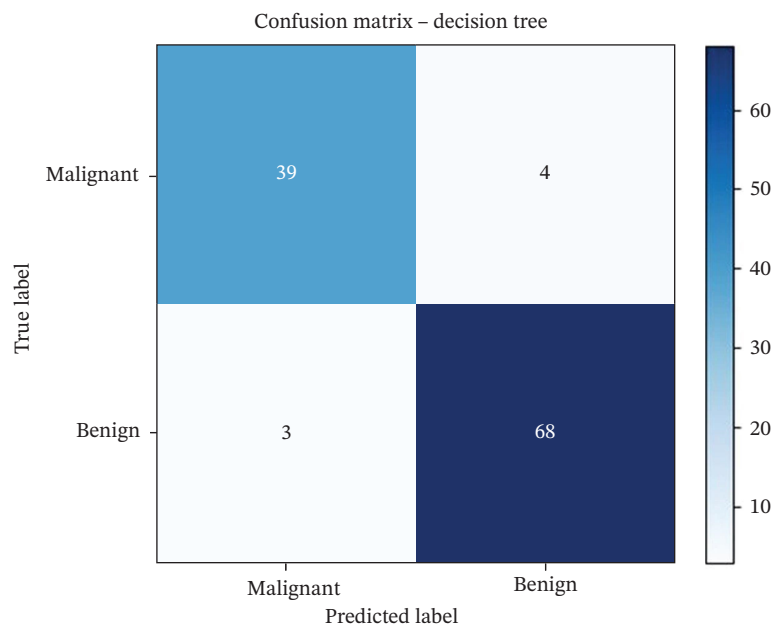


FIGURE 10 | Confusion matrix of the decision tree.

TABLE 3 | Comparison of existing studies on the IBM telecom dataset.

Authors	Dataset	Utilized algorithms	Evaluation (best model)
This study	IBM telecom dataset	Decision tree, random forest, XGBoost, logistic regression, tuned soft-voting ensemble	Accuracy: 84.73%, precision (churn): 85.20%, recall (churn): 84.07%, <i>F1</i> -score: 84.62%, ROC-AUC: 93.86%
[28]	IBM telecom customer churn dataset	Random forest, LGBM, XGBoost, logistic regression, decision tree, ANN, ensemble averaging	Accuracy: 79%, recall: 72%
[29]	IBM telco customer churn dataset	KNN, naïve Bayes, random forest, decision tree, logistic regression, artificial neural network (ANN)	ANN: 82.83% accuracy (best), KNN: 79.86% accuracy
[30]	IBM sample data sets	Hybrid two-layer flexible voting ensemble model	IBM imbalanced: accuracy 82.30%, <i>F1</i> -score 63%; IBM balanced: accuracy 76.20%, <i>F1</i> -score 77.06%

than the proposed model, suggesting that standalone deep learning or classical models without optimized combination strategies may not fully capture the dataset complexity. Similarly, the authors in [30] propose a hybrid voting-based ensemble evaluated on both imbalanced and balanced IBM datasets. While it achieves up to 82.30% accuracy on the imbalanced dataset, performance drops under balanced conditions, although *F1*-score improves to 77.06%. In contrast, the proposed model maintains stronger and more consistent performance, particularly in ROC-AUC and overall classification balance.

Overall, the comparison indicates that the proposed model achieves higher or comparable performance across most evaluation metrics when assessed against existing IBM dataset-based studies. The results also show that performance varies significantly depending on preprocessing strategies, class balancing methods, and model configuration choices used in prior work.

Table 3 compares the performance of the proposed model with existing studies that used the IBM Telco Customer Churn dataset.

5 | Conclusion and Future Works

This study demonstrates that machine learning offers a powerful approach to forecasting customer churn in the telecommunications sector, where attrition rates are notably high. By evaluating DT, XGB, RF, and LR models, we identified RF as the most effective, achieving 84.73% accuracy, 84.62% *F1*-score, and 93.86% ROC-AUC. These results highlight the model's capacity to accurately identify at-risk customers, enabling the implementation of targeted retention strategies. The findings confirm that integrating advanced predictive analytics into telecom operations can substantially reduce churn, strengthen customer relationships, and enhance competitive advantage in a saturated market. It is important to note that predictive performance improvements alone do not imply causal impact on customer retention. Realizing churn reduction requires integrating prediction models with intervention strategies, cost-sensitive decision rules, and empirical validation through controlled experiments or longitudinal deployment. This study therefore positions churn prediction as a decision-support component rather than a direct churn reduction mechanism.

Since the dataset used in this study is static and historical, model performance on unseen data from a different time period or geographic region may vary due to changes in customer behavior, market conditions, and service offerings (i.e., dataset shift or concept drift). While the current models demonstrate strong performance under the given data distribution, their generalization to new populations cannot be guaranteed without external validation. In real-world deployment, periodic retraining and validation on recent or region-specific data would be necessary to maintain performance stability and adaptability over time.

Future studies will explore advanced deep learning architectures, including temporal and sequential models, to capture long-range behavioral patterns more effectively. This study will also address the black-box nature of ensemble models by incorporating explainable AI techniques such as SHAP and permutation importance to improve transparency [31]. These methods will help provide both global and local explanations of predictions, making the models more interpretable and suitable for business decision-making [32].

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No funding was received for this study.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data supporting the findings of this study are publicly available. The primary dataset used in this study is the "Telco Customer Churn" dataset, provided through Kaggle and originally derived from IBM sample datasets. The dataset is available at <https://www.kaggle.com/datasets/blastchar/telco-customer-churn>.

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